Combined Mock Exam (Deduplicated)

This reorganized question bank combines **mock-exam-1.pdf** and **zzz.pdf**.

- Repeated questions were merged into one version.
- Unique questions from both PDFs were retained.
- Every question includes answer choices.
- The answer and reasoning appear below each question.

Section A. AI and Business Transformation

1. What is the main risk of an un-grounded LLM?

- A) It cannot write code
- B) It may hallucinate and make up answers
- C) It cannot read files
- D) It only works offline

Answer: B

Reasoning: The course defines hallucination as an LLM giving a confident but false or invented answer when it is not properly connected to tools or live data.

2. Traditional SaaS software is being challenged because AI agents can now execute many workflows dynamically.

- A) True
- B) False

Answer: A

Reasoning: The lectures argue that rigid SaaS products are under pressure because AI agents can handle many tasks more flexibly across tools and workflows.

3. What is the role of a person who manages a fleet of AI agents to run business tasks?

- A) Data Engineer
- B) Software Developer
- C) Agent Orchestrator
- D) SaaS Manager

Answer: C

Reasoning: An Agent Orchestrator is responsible for coordinating and supervising multiple AI agents to complete business tasks effectively.

4. An un-grounded LLM can accurately retrieve your real-time bank balance without any external tools.

- A) True
- B) False

Answer: B

Reasoning: A model without access to tools or live systems cannot fetch real-time personal data such as a current bank balance.

5. Which expression matches the course framework for reliable AI?

- A) Python + VS Code = Output
- B) LLM + Tool + Live Data = Grounded Action
- C) Copilot + Excel = Dashboard
- D) OpenClaw + API = Spreadsheet

Answer: B

Reasoning: The course emphasizes that AI becomes reliable when the language model is combined with tools and real data instead of operating from guesswork alone.

6. OpenClaw is best described as:

- A) A paid enterprise SaaS platform
- B) A free, local, autonomous AI agent
- C) A cloud-based coding assistant
- D) A Bloomberg terminal plugin

Answer: B

Reasoning: OpenClaw is presented as a free and local autonomous agent framework rather than a traditional paid SaaS tool.

7. Which messaging platform does OpenClaw use as its user interface?

- A) Slack
- B) Microsoft Teams
- C) Telegram, Discord, or WhatsApp
- D) Gmail

Answer: C

Reasoning: The course explains that OpenClaw uses common messaging apps as the user-facing interface layer.

8. GitHub Copilot can fully replace the need for domain knowledge in business data analysis.

- A) True
- B) False

Answer: B

Reasoning: The lectures stress that AI helps users work faster, but domain knowledge is still needed to ask the right questions and judge whether outputs make sense.

9. Fill in the blank: "AI does not _____ everyone — it _____ the gap between those who know what they are doing and those who don't."

- A) Equalize / Amplifies
- B) Help / Closes
- C) Equalize / Closes
- D) Replace / Creates

Answer: A

Reasoning: The course repeatedly argues that AI increases the advantage of skilled users rather than making everyone equally capable.

Section B. Python Basics

10. Why is Conda useful?

- A) It replaces Python
- B) It creates isolated virtual environments
- C) It is only for Jupyter markdown
- D) It removes the need for packages

Answer: B

Reasoning: Conda helps separate project environments so packages and versions do not conflict across different tasks.

11. A variable in Python is best described as:

- A) A fixed keyword
- B) A named storage location for data
- C) A chart type
- D) A file extension

Answer: B

Reasoning: Variables are names that point to values stored in memory, which lets the program reuse and modify data.

12. Which symbol is the assignment operator in Python?

- A) ==
- B) :
- C) =
- D) ->

Answer: C

Reasoning: The single equals sign assigns a value to a variable, while double equals checks equality.

13. Which variable name is invalid in Python?

- A) student_count
- B) avgtemp
- C) 2students
- D) class2

Answer: C

Reasoning: Python variable names cannot begin with a digit, so `2students` breaks the naming rules.

14. What is the output type of `type(25)`?

- A) float
- B) int
- C) str
- D) bool

Answer: B

Reasoning: The value 25 is a whole number, and Python stores whole numbers as integers.

15. In Python, the `/` operator always returns a:

- A) string
- B) boolean
- C) float
- D) list

Answer: C

Reasoning: Standard division in Python produces a floating-point result, even when both inputs are integers.

16. What is the output of the following code?

```
mylist = [1, 2, 3, 4, 5]
print(mylist[1:4])
```

- A) `[1, 2, 3]`
- B) `[2, 3, 4]`
- C) `[2, 3, 4, 5]`
- D) `[1, 2, 3, 4]`

Answer: B

Reasoning: Python slicing includes the start index and excludes the end index, so positions 1, 2, and 3 are returned.

17. Python list indexing starts at 1.

- A) True
- B) False

Answer: B

Reasoning: Python uses zero-based indexing, so the first item is at index 0.

18. Which data type represents truth values?

- A) str
- B) bool
- C) float
- D) tuple

Answer: B

Reasoning: Boolean values are `True` and `False`, and their type in Python is `bool`.

19. Which syntax is used for an f-string?

- A) `"My name is {name}"`
- B) `f"My name is {name}"`
- C) `str("My name is {name}")`
- D) `format"My name is {name}"`

Answer: B

Reasoning: An f-string must have `f` directly before the opening quotation mark to evaluate expressions inside braces.

20. An `if` statement condition in Python should end with a colon.

- A) True
- B) False

Answer: A

Reasoning: Python requires a colon at the end of control statements such as `if`, `elif`, `else`, `for`, and `while`.

21. In Python, `elif` means:

- A) end loop if
- B) else if
- C) equal if
- D) extra line if

Answer: B

Reasoning: `elif` adds another condition branch after an initial `if` statement.

22. The logical operator `and` returns `True` when:

- A) At least one condition is True
- B) Both conditions are True
- C) Both conditions are False
- D) The first condition is False

Answer: B

Reasoning: `and` is only true when every connected condition is true.

23. What does the following code do?

```
for fruit in ["apple", "banana", "cherry"]:
    print(fruit)
```

- A) Prints all three fruits at once
- B) Prints each fruit one by one on separate lines
- C) Creates a new list
- D) Returns an error

Answer: B

Reasoning: The loop runs once for each item, so each fruit is printed separately in sequence.

24. What numbers are produced by `range(1, 5)`?

- A) 1, 2, 3, 4
- B) 1, 2, 3, 4, 5
- C) 0, 1, 2, 3, 4
- D) 2, 3, 4, 5

Answer: A

Reasoning: `range(1, 5)` starts at 1 and stops before 5.

25. What does `break` do inside a loop?

- A) Skips one iteration
- B) Stops the loop completely
- C) Restarts the loop
- D) Creates an error

Answer: B

Reasoning: `break` exits the loop immediately when a chosen condition is met.

26. A `while` loop runs only once, regardless of the condition.

- A) True
- B) False

Answer: B

Reasoning: A `while` loop continues running as long as its condition stays true.

27. What is the correct syntax to define a function in Python?

- A) `function add(x, y):`
- B) `def add(x, y):`
- C) `define add(x, y):`
- D) `fun add(x, y):`

Answer: B

Reasoning: Python uses the `def` keyword to define a function.

28. List comprehension is mainly used to:

- A) delete files
- B) create a new list in one line
- C) sort dictionaries only

- D) open notebooks

Answer: B

Reasoning: List comprehension is a compact way to build a list by transforming or filtering another iterable.

29. Lists are mutable in Python.

- A) True
- B) False

Answer: A

Reasoning: Lists can be modified after creation, for example by adding, removing, or changing elements.

30. Which data structure is immutable?

- A) list
- B) dictionary
- C) tuple
- D) set

Answer: C

Reasoning: Tuples cannot be changed after they are created, which is what immutable means.

31. Which data structure does NOT allow duplicate values?

- A) List
- B) Tuple
- C) Set
- D) Dictionary value

Answer: C

Reasoning: A set automatically keeps only unique values, so duplicates are removed.

32. A dictionary stores data as:

- A) rows and columns
- B) key-value pairs
- C) indexed-only numbers
- D) sorted duplicates

Answer: B

Reasoning: Dictionaries map keys to values, which makes them useful for fast lookup by name rather than by position.

33. What is the output of the following code?

```
mydict = {"name": "John", "age": 25}
print(mydict["age"])
```

- A) `"name"`
- B) `"John"`
- C) `25`
- D) Error

Answer: C

Reasoning: The key `"age"` points to the value 25, so that value is returned.

34. What does the `append()` method do for a list?

- A) Removes the first item
- B) Adds an item to the end
- C) Sorts the list
- D) Converts to tuple

Answer: B

Reasoning: `append()` inserts one new element at the end of the existing list.

35. What does `len()` return?

- A) The maximum value
- B) The data type
- C) The number of items
- D) The average

Answer: C

Reasoning: `len()` counts how many elements are inside a sequence or collection.

36. After `import math`, what is the result of `math.sqrt(25)`?

- A) 25
- B) 10
- C) 5.0
- D) 2.5

Answer: C

Reasoning: The square root of 25 is 5, and `math.sqrt()` returns the result as a float.

Section C. Pandas, Data Loading, and EDA

37. A pandas DataFrame is best described as:

- A) A Python list
- B) An Excel spreadsheet or SQL table
- C) A plain text file
- D) A loop method

Answer: B

Reasoning: A DataFrame stores tabular data in rows and columns, which makes it similar to spreadsheets and database tables.

38. In real-world data analysis, the course recommends using _____ instead of basic file-by-file reading with `open()`.

- A) random
- B) pandas
- C) turtle
- D) math

Answer: B

Reasoning: Pandas is designed for structured tabular data and is much more efficient for real-world business datasets.

39. Which `os` function changes the working directory?

- A) `os.getcwd()`
- B) `os.chdir()`
- C) `os.listdir()`
- D) `os.path.join()`

Answer: B

Reasoning: `os.chdir()` changes the current working directory, while `os.getcwd()` only displays it.

40. Which pandas method is used to load a CSV file into a DataFrame?

- A) `pd.load_csv()`
- B) `pd.open_csv()`
- C) `pd.read_csv()`

- D) `pd.import_csv()`

Answer: C

Reasoning: `pd.read_csv()` is the standard pandas function for importing CSV data into a DataFrame.

41. By default, `df.head()` shows how many rows?

- A) 3
- B) 5
- C) 10
- D) All rows

Answer: B

Reasoning: Unless another value is specified, `head()` displays the first 5 rows.

42. Which statement about `iloc` and `loc` is correct?

- A) `iloc` uses labels and `loc` uses positions
- B) Both only use labels
- C) `iloc` uses numerical positions and `loc` uses labels or conditions
- D) Both only use positions

Answer: C

Reasoning: `iloc` is position-based indexing, while `loc` is label-based and can also be used with Boolean conditions.

43. To select only rows where the `season` column equals 2, which expression is correct?

- A) `df.loc[df["index"] == 2]`
- B) `df.loc[df["season"] == 2]`
- C) `df.loc[df["row"] == 2]`
- D) `df.loc[df["filter"] == 2]`

Answer: B

Reasoning: The filter must reference the actual column name you want to test, which is `season`.

44. What should you use when combining multiple conditions in pandas filtering?

- A) `and` only
- B) `&`, with each condition wrapped in parentheses
- C) `+`
- D) `,`

Answer: B

Reasoning: Pandas uses elementwise logical operators such as `&`, and each separate condition must be wrapped in parentheses.

45. To select only integer columns from a DataFrame, which code is correct?

- A) `df.select_rows(int)`
- B) `df.select_dtypes(include="int")`
- C) `df.filter(int)`
- D) `df.iloc["int"]`

Answer: B

Reasoning: `select_dtypes()` is designed for type-based filtering, and `include="int"` keeps integer columns.

46. In a pandas operation, what does `axis=1` refer to?

- A) Rows
- B) Values
- C) Columns
- D) Index numbers

Answer: C

Reasoning: In pandas, axis 0 is the row direction and axis 1 is the column direction.

47. Which method would you use to rename a column in a DataFrame?

- A) `.relabel()`
- B) `.rename()`
- C) `.change()`
- D) `.column_name()`

Answer: B

Reasoning: `.rename()` is the pandas method used to change row or column labels.

48. In `rename()`, which axis value targets columns?

- A) `axis=0`
- B) `axis=1`
- C) `axis=-1`
- D) `axis=None`

Answer: B

Reasoning: `axis=1` tells pandas to apply the renaming to columns rather than rows.

49. To save a CSV file without writing the index column, which argument should be used?

- A) `header=False`
- B) `axis=1`
- C) `index=False`
- D) `drop=True`

Answer: C

Reasoning: `index=False` prevents pandas from exporting the row index as an extra column.

50. What information does `df.info()` mainly provide?

- A) Only the mean of numeric columns
- B) Rows, columns, non-null counts, and data types
- C) Only missing values
- D) Only unique values

Answer: B

Reasoning: `info()` is a structural summary of the DataFrame, not a statistical summary.

51. What does `df.describe()` mainly summarize by default?

- A) Only text columns
- B) Only index labels
- C) Numerical columns
- D) Only missing rows

Answer: C

Reasoning: By default, `describe()` returns summary statistics for numeric columns such as mean, standard deviation, and quartiles.

52. What does `value_counts()` show for a column?

- A) The column type
- B) Frequency of each unique value
- C) The mean and median
- D) Correlation coefficients

Answer: B

Reasoning: `value_counts()` counts how often each distinct value appears in a Series.

53. What does `unique()` return?

- A) Distinct values without counts
- B) Sorted values with percentages
- C) Only the most common value
- D) Only missing values

Answer: A

Reasoning: `unique()` lists the different observed values but does not count them.

54. Why would you use `pd.to_datetime(..., errors="coerce")`?

- A) To sort by date automatically
- B) To turn invalid date values into missing values instead of stopping with an error
- C) To convert dates into integers
- D) To drop all date columns

Answer: B

Reasoning: `errors="coerce"` converts bad date strings to missing values so the rest of the code can continue running.

55. Which pandas method is used to detect missing values in a column?

- A) `.missing()`
- B) `.isnull()`
- C) `.isna()`
- D) Both B and C are correct

Answer: D

Reasoning: `.isnull()` and `.isna()` are equivalent in pandas and both identify missing values.

56. Deleting rows with missing values is always the best solution, regardless of dataset size.

- A) True
- B) False

Answer: B

Reasoning: Removing missing rows can throw away too much information, especially in smaller datasets or when missingness is systematic.

57. What does `dropna()` do by default?

- A) Drops all columns with missing values
- B) Drops any row that has at least one missing value
- C) Replaces `NaN` with 0
- D) Returns a count of missing values

Answer: B

Reasoning: The default behavior of `dropna()` is to remove rows containing missing values.

58. Which method is used to replace missing values with a mean or median?

- A) `fillna()`
- B) `clip()`
- C) `agg()`
- D) `merge()`

Answer: A

Reasoning: `fillna()` is used to insert a chosen replacement value such as a column mean or median.

59. To replace missing values in a column with the column mean, which code completes the blank correctly?

```
df["column"].fillna(df["column"].________())
```

- A) `median`
- B) `average`
- C) `mean`
- D) Either A or C

Answer: C

Reasoning: The blank asks specifically for replacement with the column mean, so the correct method is `mean()`.

60. What is the main purpose of `groupby(...).agg(...)`?

- A) To rename variables
- B) To compute selected summary statistics by group
- C) To create missing values
- D) To reverse rows

Answer: B

Reasoning: `groupby()` splits the data into categories and `agg()` calculates summary statistics such as mean, count, or standard deviation for each group.

61. In the `.groupby()` function, what does `.agg()` stand for?

- A) Aggregate
- B) Align
- C) Argument
- D) Adjust

Answer: A

Reasoning: `.agg()` is short for aggregate because it summarizes grouped data using one or more functions.

62. Which chart type is best for spotting individual extreme outlier dots in your data?

- A) Histogram
- B) Line chart
- C) Boxplot
- D) Pie chart

Answer: C

Reasoning: A boxplot clearly displays outliers as points beyond the whiskers, making extremes easier to see.

63. What is winsorization?

- A) Deleting all outliers from the dataset
- B) Capping extreme values at defined quantile limits instead of removing them
- C) Replacing missing values with the mean
- D) Sorting the data from smallest to largest

Answer: B

Reasoning: Winsorization reduces the influence of extreme values while keeping the observations in the dataset.

64. In quantile-based winsorization, which pair is used for top 1 percent and bottom 1 percent capping?

- A) 0.90 and 0.10
- B) 0.95 and 0.05
- C) 0.99 and 0.01
- D) 1.00 and 0.00

Answer: C

Reasoning: Top 1 percent and bottom 1 percent capping uses the 99th and 1st percentiles.

65. The Z-score method identifies outliers based on how many standard deviations a value is from the mean.

- A) True
- B) False

Answer: A

Reasoning: Z-scores measure distance from the mean in standard deviation units, which is why they are used for outlier detection.

66. What is the goal of z-score standardization?

- A) Convert data to text
- B) Rescale data to mean 0 and standard deviation 1
- C) Remove all outliers
- D) Turn floats into integers

Answer: B

Reasoning: Standardization makes variables comparable by centering them at 0 and scaling them to a standard deviation of 1.

67. Forward fill uses the next available value to replace a missing value in time-series data.

- A) True
- B) False

Answer: B

Reasoning: Forward fill uses the previous observed value, not the next one. Using the next value would be backward fill.

Section D. Correlation and Regression

68. What is the range of a Pearson correlation coefficient (r)?

- A) 0 to 1
- B) -100 to 100
- C) -1 to 1
- D) 0 to 100

Answer: C

Reasoning: Correlation strength and direction are always measured on a scale from -1 to 1.

69. Correlation implies causation.

- A) True
- B) False

Answer: B

Reasoning: Two variables may move together for many reasons, so correlation alone does not prove that one causes the other.

70. Correlation mainly measures _____, while slope mainly measures _____.

- A) steepness; color
- B) tightness of relationship; rate of change
- C) prediction; missingness
- D) category; rank

Answer: B

Reasoning: Correlation tells how closely two variables move together, while slope tells how much Y changes for a one-unit increase in X.

71. A steep regression line always means a strong correlation between two variables.

- A) True
- B) False

Answer: B

Reasoning: A steep slope only shows the rate of change in units, not how tightly the data points follow the line.

72. Which correlation method should you use when your data is skewed or contains significant outliers?

- A) Pearson
- B) Spearman
- C) OLS
- D) Z-score

Answer: B

Reasoning: Spearman uses ranks instead of raw values, so it is less sensitive to skewness and outliers.

73. One major reason to examine a correlation matrix before regression is to detect _____ among independent variables.

- A) autocorrect
- B) multicollinearity
- C) indexing
- D) winsorization

Answer: B

Reasoning: A correlation matrix helps identify predictors that are highly correlated with each other, which can destabilize regression estimates.

74. Simple Linear Regression uses _____ independent variable and _____ dependent variable.

- A) one; one
- B) two; one
- C) one; two
- D) many; none

Answer: A

Reasoning: Simple Linear Regression models the relationship between one predictor and one outcome variable.

75. In regression, the _____ tells you how much Y changes for every 1-unit increase in X.

- A) R-squared
- B) P-value
- C) Intercept
- D) Coefficient (Beta)

Answer: D

Reasoning: The coefficient, or slope, is the term that captures the exact rate of change between X and Y.

76. A t-statistic larger than about 2 in absolute value generally suggests:

- A) very weak evidence
- B) the signal is stronger than the noise
- C) the model should be deleted
- D) no relationship at all

Answer: B

Reasoning: A large absolute t-statistic means the estimated effect is large relative to its uncertainty.

77. A p-value of 0.03 means:

- A) There is a 97% chance the result is random
- B) There is a 3% chance the result happened by pure luck, so it is statistically significant
- C) The model explains 3% of the variance
- D) The coefficient is 0.03

Answer: B

Reasoning: In course-style interpretation, a p-value below 0.05 is treated as statistically significant, meaning the observed result would be unlikely under the null hypothesis.

78. A p-value must be below _____ to be considered statistically significant at the standard threshold.

- A) 0.10
- B) 0.50
- C) 0.05
- D) 0.01

Answer: C

Reasoning: The standard threshold used in the course is 0.05.

79. What does R-squared (R^2) measure in a regression model?

- A) The slope of the regression line
- B) How much variance in Y is explained by the model
- C) The number of observations used
- D) The standard deviation of residuals

Answer: B

Reasoning: `R^2` measures the share of variation in the dependent variable that the model explains.

80. What does the "L" in the L.I.N.E. regression assumptions stand for?

- A) Large sample size
- B) Linearity
- C) Low variance
- D) Log transformation

Answer: B

Reasoning: Linearity means the relationship between X and Y should be approximately straight-line for standard linear regression interpretation.

81. In the L.I.N.E. assumptions for OLS, the letter "E" stands for:

- A) Extra variables
- B) Equal variance
- C) Expected value
- D) Extrapolation

Answer: B

Reasoning: Equal variance means the spread of residuals should remain reasonably constant across fitted values.

82. Multicollinearity occurs when two independent variables are highly correlated with each other, which causes problems in regression.

- A) True
- B) False

Answer: A

Reasoning: Highly correlated predictors make it harder to isolate each variable's individual effect and can make coefficients unstable.

83. Which statement about regression is correct?

- A) A statistically significant coefficient always proves causation
- B) Regression can safely predict far outside the observed data range
- C) Outliers can strongly distort a regression line
- D) Regression does not require a research question

Answer: C

Reasoning: OLS regression is sensitive to extreme values, so outliers can pull the fitted line and distort the estimated relationship.